

Section 5.4

46. a. We know $\mu=12$ and $\sigma=0.04$, $n=16$.

Thus $E(\bar{X}) = \mu_{\bar{X}} = \mu = 12$, i.e., $\bar{X}$ centers at 12.

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = 0.01$$

b. We know $\mu=12$, $\sigma=0.04$, $n=64$.

Thus $E(\bar{X}) = \mu_{\bar{X}} = \mu = 12$, i.e., $\bar{X}$ centers at 12.

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = 0.005$$

c. For $n=16$, $P(11.99 \leq \bar{X} \leq 12.01)$

$$= P\left(\frac{11.99-12}{0.01} \leq Z \leq \frac{12.01-12}{0.01}\right)$$

$$= P(-1 \leq Z \leq 1)$$

For $n=64$, $P(11.99 \leq \bar{X} \leq 12.01)$

$$= P\left(\frac{11.99-12}{0.005} \leq Z \leq \frac{12.01-12}{0.005}\right)$$

$$= P(-2 \leq Z \leq 2)$$

By the property of normal distribution,

$P(-2 \leq Z \leq 2) > P(-1 \leq Z \leq 1)$. Thus part b is more likely.

51. For the population. $\mu=10$ and $\sigma=2$.

Let $\bar{X}_1$ denote the mean value of the first day.

Then $n_1=5$. Thus $E(\bar{X}_1)=\mu_{\bar{X}_1}=\mu=10$, $\sigma_{\bar{X}_1}=\frac{\sigma}{\sqrt{n_1}}=0.894$

$$\text{Thus } P(\bar{X}_1 \leq 11) = P\left(Z \leq \frac{11-10}{0.894}\right) = P(Z \leq 1.12)$$

$$= \Phi(1.12)$$

$$= 0.8686$$

Similarly, $n_2=6$. Thus $E(\bar{X}_2)=\mu_{\bar{X}_2}=\mu=10$, $\sigma_{\bar{X}_2}=\frac{\sigma}{\sqrt{n_2}}=0.816$

$$\text{Thus } P(\bar{X}_2 \leq 11) = P\left(Z \leq \frac{11-10}{0.816}\right) = P(Z \leq 1.23)$$

$$= \Phi(1.23)$$

$$= 0.8907.$$

$$\text{Thus } P(\bar{X}_1 \leq 11 \text{ and } \bar{X}_2 \leq 11)$$

$$= P(\bar{X}_1 \leq 11) \cdot P(\bar{X}_2 \leq 11)$$

$$= 0.774$$

53. a. Since μ is large, the number of parking tickets has approximately normal distribution.

$\sigma = \sqrt{\mu} = 2.071$. Let X denote the tickets number on a certain day.

$$\begin{aligned}
 \text{Thus } P(35 \leq X \leq 70) &= P\left(\frac{35-50-0.5}{7.071} \leq Z \leq \frac{70-50+0.5}{7.071}\right) \\
 &= P(-2.19 \leq Z \leq 2.90) \\
 &= \Phi(2.90) - \Phi(-2.19) \\
 &= 0.9981 - 0.0143 \\
 &= 0.9838
 \end{aligned}$$

b. For $T_0 = \sum X$, $n=5$.

$$\text{Thus } E(T_0) = n\mu = 250. \quad \sigma_{T_0} = \sqrt{n}\sigma = 15.811$$

$$\text{Thus } P(225 \leq T_0 \leq 275)$$

$$= P\left(\frac{225-250-0.5}{15.811} \leq Z \leq \frac{275-250+0.5}{15.811}\right)$$

$$= P(-1.61 \leq Z \leq 1.61)$$

$$= \Phi(1.61) - \Phi(-1.61)$$

$$= 0.9463 - 0.0537$$

$$= 0.8926$$

Section 5.5

58. a. Let V denote the total volume.

$$\text{Then } V = 27X_1 + 125X_2 + 512X_3$$

$$\begin{aligned}\text{Thus } E(V) &= 27E(X_1) + 125E(X_2) + 512E(X_3) \\ &= 87850\end{aligned}$$

Since they are independent,

$$\begin{aligned}V(V) &= 27^2 V(X_1) + 125^2 \cdot V(X_2) + 512^2 \cdot V(X_3) \\ &= 1910016\end{aligned}$$

b. For $E(V)$, it is still correct since

$E(V) = \sum a_i E(X_i)$ is correct for any random variables.

For $V(V)$, it is not necessarily correct

since $V(V) = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)$ for dependent variables.

70. a. $E(Y_i) = 0 \times 0.5 + 1 \times 0.5 = 0.5$

$$E(W) = \sum_{i=1}^n i \cdot E(Y_i) = 0.5 \sum_{i=1}^n i = \frac{n(n+1)}{4}$$

b. $V(Y_i) = (0-0.5)^2 \cdot 0.5 + (1-0.5)^2 \cdot 0.5 = 0.25$

$$V(W) = \sum_{i=1}^n i^2 \cdot V(Y_i) = 0.25 \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{24}$$

73. a. Since $n_1 = 40 > 30$, $n_2 = 35 > 30$, we can use CLT.

$\bar{X}, \bar{Y}$ all have approximately normal distribution

For $\bar{X}$, $E(\bar{X}) = \mu_{\bar{X}} = \mu = 105$, $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n_1} = 1.60$

Thus $\bar{X} \sim N(105, 1.6)$

For $\bar{Y}$, $E(\bar{Y}) = \mu_{\bar{Y}} = \mu = 100$, $\sigma_{\bar{Y}}^2 = \frac{\sigma^2}{n_2} = 1.03$

Thus $\bar{Y} \sim N(100, 1.03)$

b. Since $\bar{X} - \bar{Y}$ is a linear combination of $\bar{X}$ and $\bar{Y}$, $\bar{X}, \bar{Y}$ are all approximately normal distribution

Thus $\bar{X} - \bar{Y}$ has approximately normal distribution.

c. $E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = 105 - 100 = 5$.

$$\sigma_{\bar{X} - \bar{Y}} = \sqrt{\sigma_{\bar{X}}^2 + \sigma_{\bar{Y}}^2} = 1.622$$

$$\begin{aligned}
 \text{Thus } P(-1 \leq \bar{Y} - \bar{Y} \leq 1) &= P\left(\frac{-1-5}{1.622} \leq Z \leq \frac{1-5}{1.622}\right) \\
 &= P(-3.70 \leq Z \leq -2.47) \\
 &= \Phi(-2.47) - \Phi(-3.70) \\
 &= 0.0068 - 0.0001 = 0.0067
 \end{aligned}$$

$$\begin{aligned}
 \text{d. } P(\bar{X} - \bar{Y} \geq 10) &= P\left(Z \geq \frac{10-5}{1.622}\right) = P(Z \geq 3.08) \\
 &= 1 - \Phi(3.08) \\
 &= 1 - 0.9990 \\
 &= 0.0010
 \end{aligned}$$

Since $0.0010 \approx 0$, that is, the probability is extremely small. Thus we should doubt if we actually observed it.

